06 Chemical Synthesis — Problem-solving approach

Basis: sub-questions tagged SYN in Section One (multiple choice), Section Two and Section Three of the 2016–2025 Chemistry ATAR past papers, 5 patterns in total, 61 tag occurrences altogether (18 in Section One multiple-choice questions, 43 in Section Two and Section Three). Marking points are taken from the key in the marking criteria, paraphrased rather than copied verbatim; the original papers and marking files remain in the local ignored directory. Past-paper numbering: `2016M-Q23` = 2016 Section One (multiple choice) Question 23; `2021E-Q38` = 2021 Section Three Question 38 (whole question, no sub-parts); `2023E-Q39(bii)` = 2023 Section Three Question 39 (b)(ii). Pattern definitions are taken from `tools/catalogue_CHE.json`; tags are draft (draft_for_review), and this file has not been independently reviewed. ★ marker: ranked by each pattern's total marks (INDEX.md), taking roughly the top quarter (1 of 5 patterns); this file is SYN4 (67 marks). SYN2 (65 marks) is close behind and remains an important pattern for this topic. Cross-topic overlaps: Pattern 1 overlaps with ORG5–ORG7; Pattern 2 differs from EQ4 — questions that only use Le Châtelier's principle to predict a shift belong to EQ4; Pattern 4 overlaps with ORG4; Pattern 5 overlaps with PRO7 (2020E-Q40(f) counts towards both SYN3 and PRO7, see Pattern 7 in `05_Polymers_Proteins_解题思路.md`).

Marking panel notes

- Long synthesis questions are often left blank: 2020 report — 2020S-Q33 (synthesising ethyl pentanoate from ethene, pentanal or pentan-1-ol, involving SYN1, ORG5, ORG6, ORG7) averaged 39.53%, the hardest Section Two question on that paper, and many candidates left it blank.
- Combined rate-and-yield questions are the hardest: 2021 report — 2021E-Q38 (Contact process, combining collision theory and chemical equilibrium) averaged 35.5%. 2024 report — 2024E-Q36(a) (requiring candidates to "outline and justify" the optimum conditions) was the hardest sub-question.
- Comparing soap and detergent structures: the 2024 report states that 2024S-Q31 (similarities and differences in the structures of soap and detergent, plus a micelle diagram) was the best-performed question in Section Two.
- One paragraph must cover both aspects: the key for industrial-conditions questions generally requires an explanation of both rate (collision theory) and yield (Le Châtelier's principle). Writing only one aspect will not earn full marks.

Pattern 1: Synthesis routes, industrial reactions and conditions

Identification: the question asks you to "design / complete a synthesis route", "use equations to show how … can be produced", "write an equation for the fermentation / hydration process and state the conditions", or "combine these equations to produce an overall equation".

Steps

1. List the starting materials and the target product, then write out each step: reactants → products, reagents or catalysts, conditions, and reaction type (hydration, oxidation, esterification, fermentation, etc.).
2. For industrial reactions, give all the conditions: catalyst (H^+, such as H_3PO_4, H_2SO_4; or an enzyme, such as zymase, lipase), temperature, pressure, and phase (water taking part as $H_2O(g)$).
3. Write equations using condensed structures; balance them; where needed, write the catalyst above the arrow.
4. For a multi-step synthesis, if the question asks for an overall equation, cancel the intermediates from each step, write the reactants and products of the overall reaction, then balance it.
5. When choosing starting materials, check that each step is feasible: for example, hydration of ethene gives ethanol, oxidation of pentanal or pentan-1-ol gives pentanoic acid, and esterification of ethanol with pentanoic acid gives ethyl pentanoate.

Marking points (typical wording in the key): correct reactants and products for each step (2021E-Q39(a) key: the ethene hydration step is worth 4 marks — 1 each for reactants, products, acid catalyst and $H_2O(g)$; the biodiesel step is worth 4 marks — 1 each for reactants, products, balancing and lipase); for ethene hydration, water as gaseous $H_2O(g)$ and an acid catalyst specified (2021E-Q39(a)); fermentation products must include CO_2 (2022S-Q33(a): 2 C_2H_5OH + 2 CO_2, balanced, yeast or zymase as catalyst, and any one of temperature,

pH or pressure); the hydration conditions (any one of 300 °C, about 6000–7000 kPa, or water as a gas; 2022S-Q33(a)); 2019S-Q29(a) combining two equations to give an overall equation — 1 mark each for products, reactants and balancing; 2018S-Q30(c) the two reagents for step 3, synthesising (C) from (B): acidified dichromate or permanganate (or sulfuric acid plus one of these), with the key accepting either of two answers.

Common errors (exam reports)

- The 2020S-Q33 (ethyl pentanoate) key required students to carry out the ethanol synthesis and the pentanoic acid synthesis in two separate reaction vessels, then esterify. The report states this question averaged 39.53%, the hardest on the paper, and many candidates did not attempt it.
- Omitting CO_2 from the fermentation equation (the keys for both 2022S-Q33(a) and 2023E-Q39(aii) require correct products and balancing).
- Omitting the catalyst or the phase of the water in ethene hydration (the 2021E-Q39(a) key requires $H_2O(g)$).
- 2023E-Q39(aii) instructed candidates to use condensed structures; answers should be written in the representation specified in the question.
- MC questions give a process flow chart and ask you to name reaction X, gas Y or process Z (2016M-Q23 to Q25); first decide the reaction type (e.g. esterification, fermentation or fractional distillation), then match it to the gas and step in the flow chart.

Past questions: 2016M-Q23, 2016M-Q24, 2016M-Q25, 2016S-Q30(a), 2016S-Q30(b), 2018S-Q30(c), 2019S-Q29(a), 2020S-Q33, 2021E-Q39(a), 2022S-Q33(a), 2023E-Q39(a)(ii)

Pattern 2: Industrial conditions: compromise between rate and yield

Identification: The question asks "state the conditions that would optimise the rate / yield", "state a compromise", "explain the effect on rate and yield", "justify the temperature and pressure", "explain why excess water increases the rate and yield"; the question often gives an industrial process (Haber process, Contact process, hydration, biodiesel, soap production, etc.).

Steps (matching the convention for this topic: one explanation covers both rate and yield)

1. State the thermal effect and the change in moles of gas: whether the forward reaction is exothermic or endothermic; whether the total moles of gaseous reactants or products is greater.
2. Rate (collision theory):
    - Increasing temperature → average kinetic energy of molecules increases → collision frequency increases, and the proportion of collisions with energy greater than or equal to the activation energy (E_a) increases → rate increases;
    - Increasing pressure (gases) → number of particles per unit volume increases → collision frequency increases → rate increases;
    - A catalyst lowers E_a, provides an alternative reaction pathway, and increases the proportion of effective collisions; it speeds up the forward and reverse reactions equally and does not change the equilibrium position.
3. Yield (Le Châtelier's principle):
    - Exothermic reaction: lowering temperature favours the forward reaction; endothermic reaction: raising temperature favours the forward reaction;
    - Direction with fewer gas molecules: increasing pressure favours the forward reaction;
    - Increasing reactant concentration or removing product favours the forward reaction.
4. Compromise: The temperature requirements for rate and yield are usually opposite, so a moderate temperature is chosen; pressure is limited by equipment and cost; a catalyst does not change yield but can lower the required temperature and pressure, improving economic viability.
5. State the conclusion: State the final chosen conditions and explain that they are a compromise

between rate and yield.

Marking points (typical key wording):

- 2017S-Q30(a): Conditions for rate (high temperature, high pressure, catalyst, at least two) and conditions for yield (low temperature, high pressure, removing product, at least two).
- 2017S-Q30(b) (6 marks): Identify one condition that requires compromise (temperature), explain its effect on rate (collision theory) and yield (equilibrium) separately, and explain why the compromise is made (total 6 marks: identify compromise condition 1 mark; rate 1–2 marks; yield 1–2 marks; explain reason for compromise 1 mark).
- 2018E-Q38(a) (7 marks): Temperature paragraph (high temperature gives a fast rate; the reaction is exothermic, so low temperature favours yield; therefore use a moderate temperature, about 300–400 °C); pressure paragraph (high pressure increases both rate and yield); economic paragraph (high temperature and high pressure are costly, a catalyst allows lower conditions to be used).
- 2021E-Q38 (12 marks): Rate (6 marks): increasing temperature increases average kinetic energy and the proportion of effective collisions; a catalyst lowers E_a and speeds up the forward and reverse reactions equally; high pressure increases particle number density. Equilibrium (3 marks): for the second step only, high temperature is unfavourable for yield (reverse reaction is endothermic), high pressure favours the forward reaction (fewer gas molecules). Compromise (2 marks): compromise on temperature, and pressure is limited by cost and safety.
- 2023E-Q35(a) (8 marks): Use collision theory only to explain catalyst (1–2 marks), temperature (1–3 marks) and pressure (1–3 marks). This question does not require an equilibrium discussion.
- 2023E-Q39(bii) (5 marks): High pressure increases both rate and yield (direction with fewer gas molecules); high temperature increases rate, and for an endothermic reaction favours product, but a moderate temperature balances economic and safety considerations.
- 2024E-Q36(a) (6 marks): Rate (high temperature, catalyst, increased concentration); yield (in an endothermic reaction high temperature favours the forward reaction, increasing reactant concentration, removing product).
- 2025E-Q37(b) (3 marks): Excess water increases collision frequency, speeding up the forward rate; adding water shifts the equilibrium forward, increasing yield.

Common errors (examiner reports)

- Writing only about rate and not yield; or only about yield and not rate. The keys for this pattern almost always require both.
- Saying a catalyst can increase yield. The keys for 2021E-Q39(e) and 2021E-Q38 both state clearly: a catalyst speeds up the forward and reverse reactions equally, does not change the equilibrium position, and yield is unchanged.
- Saying "high pressure increases yield" without giving the reason. The correct reason is that the direction with fewer gas molecules favours the forward shift in equilibrium.
- Rate explanations that only say "high temperature makes the reaction faster" and do not say "the proportion of collisions with energy greater than or equal to E_a increases". The keys for 2023E-Q35(a) and 2021E-Q38 both require this.
- 2021 report: 2021E-Q38 had an average mark of 35.5%, making it a challenging question on that paper. 2024 report: the "outline and justify" in 2024E-Q36(a) was the hardest sub-question.
- 2019E-Q37(e) predicts the optimal conditions for soap production (high temperature, high concentration of NaOH, stirring or removing product); stirring only increases surface area and rate, not yield; removing product is what affects yield.
- MC questions ask for the reason behind industrial conditions or how a condition affects rate and equilibrium yield (e.g. 2018M-Q20, 2018M-Q23, 2025M-Q5); a catalyst speeds up the rate but does not change the equilibrium yield.

Past questions: 2016E-Q39(b), 2017S-Q30(a), 2017S-Q30(b), 2018M-Q20, 2018M-Q23, 2018E-Q38(a), 2019S-Q29(b), 2019E-Q37(e), 2021E-Q38, 2023E-Q35(a), 2023E-Q39(b)(ii), 2024E-Q36(a), 2025M-Q5, 2025E-Q37(b)

Pattern 3: Green chemistry and process evaluation

Identification: The question asks "which process is preferred / more common", "explain why X is a greener process", "state reasons why fermentation is more common than hydration", "which catalyst is industrially preferred".

Steps

1. Determine the evaluation dimensions: whether the raw material is renewable, energy consumption (temperature and pressure), waste and by-products, equipment cost, safety, whether the catalyst can be reused, and environmental impact.
2. Use a comparative sentence for every reason: "A uses … whereas B uses …".
3. Reasons must be specific, not just "cheaper" or "better".
4. When choosing a catalyst, list its advantages (e.g. avoids soap formation, low energy consumption, reusable, low temperature, low toxicity), choose any three.

Marking points (typical key wording):

- 2020E-Q40(g) (2 marks): Compare biodiesel and fossil fuel diesel from two different angles, e.g. renewable raw material vs fossil fuels taking millions of years to form; production does not require high temperature and expensive, complex equipment, and has a lower carbon footprint.
- 2023E-Q39(c) (3 marks): Choose any three: fermentation requires less energy; lower cost; "greener"; uses a renewable raw material (sugar), whereas ethylene hydration uses petrochemical raw materials.
- 2020E-Q40(f) (3 marks, combined with PRO7): Lipase is industrially preferred, choose any three reasons (see PRO7).

Common errors (examiner reports)

- Writing only "greener" or "cheaper" without giving the basis for comparison.
- Confusing "renewable" with "cheap".
- The reports do not comment separately on this pattern; evidence is limited (only 3 parts).
- MC questions ask which feature is least likely to belong to green chemistry (e.g. 2021M-Q6); the principles include renewable feedstocks, less waste and fewer toxic solvents, and energy-intensive processes usually do not fit.

Past questions: 2017M-Q11, 2020E-Q40(f), 2020E-Q40(g), 2021M-Q6, 2023E-Q39(c)

Pattern 4: Soaps and Detergents ★

Identification: The question stem contains "saponification", "soap", "detergent", "micelle", "hard water", "scum", "carboxylate", "sulfonate"; or requires drawing a soap structure, labelling a micelle diagram, or writing a saponification or precipitation equation.

Steps

1. Saponification: triglyceride + 3 NaOH → glycerol + 3 $RCOO^-Na^+$ (soap). When writing the equation, include all reactants and products; glycerol is one of the products.
2. Soap structure: long hydrocarbon chain (non-polar tail, hydrophobic) + carboxylate ion head (COO^-, hydrophilic, charged). When drawing the structure, label both parts.
3. Cleaning mechanism (micelle): ionic heads face outwards, contacting water (ion-dipole forces); hydrocarbon chains face inwards, enclosing grease/oil (dispersion forces); the micelle suspends oil/grease, which is removed by agitation.
4. Orientation of water molecules: the H of water molecules points towards the COO^- end (ion-dipole forces).
5. Hard water: Ca^{2+} and Mg^{2+} form insoluble calcium (magnesium) salts with $RCOO^-$, i.e. scum; the lost surface-active ions cannot form micelles, so cleaning effectiveness decreases.

6. Synthetic detergents: the head is a sulfonate group ($-SO_3^-$); its calcium salt is soluble, so it remains effective in hard water.
7. Comparing soaps and detergents: similarities are that both have a long non-polar hydrocarbon chain and an ionic head; differences are that the heads are carboxylate and sulfonate respectively.

Marking points (typical key wording):

- Saponification equation: reactants NaOH and triglyceride, products glycerol and sodium salt, balanced (2024E-Q37(ai): missing reactant (NaOH) and product (sodium stearate) together 1–2 marks, balancing 1 mark, total 3 marks).
- 2017E-Q39(d): the 4 product structures of saponification 1 mark each (3 soap structures and glycerol).
- 2017E-Q39(ai) (5 marks): complete the hydrocarbon chain (add the CH_3 end group); label COO^- (any of Na^+, Li^+, K^+); label the long hydrocarbon chain tail as non-polar (hydrophobic); label the ionic head as hydrophilic and charged; draw water molecules with H pointing towards COO^-.
- 2019E-Q37(a) (7 marks): dispersion forces between tails and oil/grease; ion-dipole forces between heads and water; micelles suspend oil/grease, which can be removed by agitation; draw a clearly labelled diagram (both interactions: tail–oil/grease and head–water).
- 2024S-Q31(b): in the micelle diagram, W (outside the micelle), X (inside the micelle), Y (inside the micelle, where dispersion forces occur: between hydrocarbon chains or between hydrocarbon chains and oil/grease), Z (indicates the hydrophilic head group) 1 mark each, total 4 marks.
- 2022E-Q36(c) (3 marks): in hard water, the cleaning action of soap decreases while detergent remains effective; hard water contains Ca^{2+} or Mg^{2+}; soap calcium salts are insoluble, detergent calcium salts are soluble.
- 2019E-Q37(b) (4 marks): detergents do not form precipitates with ions in hard water; after soap forms a precipitate (scum), fewer ions are available to form micelles; equation has correct reactants and products, balanced (carbon chain length greater than 12 carbons, key writes $n > 12$).
- 2025E-Q35(c) (4 marks): scum is a precipitate of surface-active ions with Ca^{2+}; the precipitate reduces available surface-active ions, so more soap is needed; equation is correct.
- 2025E-Q35(e) (3 marks): the calcium salt of the detergent sulfonate is soluble and does not precipitate, so the Ca^{2+} concentration in the water is higher.

Common errors (exam reports)

- Drawing the micelle with heads facing inwards and tails facing outwards; both 2024S-Q31(b) and 2025E-Q35(b) require the correct orientation to be labelled.
- Drawing the soap tail as ionic and the head as non-polar.
- Describing the hard-water problem as "soap reacts with water" rather than "soap precipitates with Ca^{2+} or Mg^{2+}".
- 2025E-Q35(a) forms soap from caprylic acid and NaOH; this is a carboxylic acid neutralisation reaction ($RCOOH + NaOH \rightarrow RCOO^-Na^+ + H_2O$), different from saponification of triglycerides. The key requires both NaOH as a reactant and water as a product to be written.
- Report: the 2024 report lists 2024S-Q31 (similarities and differences between soap and detergent structures and the micelle diagram) as a soap topic, but the hardest questions named in that report are Q28 (solubility) and Q30 (ocean acidification), which are not part of this document.
- MC questions ask about soap and detergent structures, saponification or micelle formation (e.g. 2017M-Q21, 2018M-Q24, 2020M-Q22, 2020M-Q23, 2022M-Q4, 2023M-Q25); in hard water soap forms a precipitate, whereas synthetic detergents generally do not.

Past questions: 2017M-Q21, 2017E-Q39(a)(i), 2017E-Q39(b), 2017E-Q39(d), 2018M-Q24, 2019E-Q37(a), 2019E-Q37(b), 2019E-Q37(d), 2020M-Q9, 2020M-Q22, 2020M-Q23, 2020E-Q40(e)(i), 2022M-Q4, 2022E-Q36(a), 2022E-Q36(b), 2022E-Q36(c), 2023M-Q25, 2024S-Q31(a), 2024S-Q31(b), 2024E-Q37(a)(i), 2025E-Q35(a), 2025E-Q35(b), 2025E-Q35(c), 2025E-Q35(e)

Pattern 5: Biodiesel and Transesterification

Identification: The question stem contains "transesterification", "biodiesel", "triglyceride + ethanol / methanol", "base-catalysed vs lipase-catalysed"; or requires writing the equation for biodiesel, or comparing the two catalysis methods.

Steps

1. Write the equation: triglyceride + 3 ROH (methanol or ethanol) ⇌ 3 fatty acid alkyl esters (biodiesel) + glycerol. Write the catalyst above the arrow (base, e.g. NaOH; or lipase).
2. Check products: glycerol must be listed as a product; biodiesel is written as R–COOR′ (alkyl ester, e.g. methyl or ethyl ester).
3. Compare catalysis methods: base catalysis (NaOH) reacts with free fatty acids to form soap, reducing yield and purity; it is highly corrosive, requires high temperature, and increases the cost of subsequent purification. Lipase catalysis is mild, does not form soap, gives high product purity, makes separation from glycerol easy, and lipase is biodegradable.
4. Comparison sentence structure: "The lipase-catalysed method … whereas the base-catalysed method …".

Marking points (typical key wording):

- 2020E-Q40(a) (3 marks): glycerol as a product; correct biodiesel structure; balanced equation.
- 2024E-Q37(aii) (4 marks): reactant (methanol or ethanol) and product (methyl or ethyl stearate) 1–2 marks each; catalyst (base, NaOH) written beside the equilibrium arrow; balanced.
- 2016S-Q28(b) (3 marks): 1 mark per point, maximum 3 marks; must clearly state which catalysis method, and each point must be a comparison. Typical comparison points include: lipase method has higher yield (no soap produced); higher purity; safer (no need for corrosive strong base, no high-temperature risk); lower environmental impact (lipase is biodegradable); lower total cost (simpler glycerol purification).
- 2021E-Q39(a) (8 marks, combined with Pattern 1): lipase as catalyst, written above the arrow; correct reactants and products; balanced.

Common errors (exam reports)

- Omitting glycerol as a product and writing only the three esters.
- Catalyst does not match the method: writing NaOH for lipase, or treating NaOH as an enzyme. The 2024E-Q37(aii) key requires the base catalyst NaOH to be written beside the equilibrium arrow (the base-catalysed method), to distinguish it from the lipase method in the question stem.
- Writing unsupported comparisons such as "lipase is cheaper" without identifying the specific difference from the base-catalysed method.
- The marking for 2016S-Q28(b) requires each point to state which method it refers to; otherwise no mark is awarded.
- MC questions ask about the reactants, catalysts or composition of biodiesel in transesterification (e.g. 2016M-Q5, 2019M-Q11, 2019M-Q13); triglycerides react with methanol to give methyl esters and glycerol, and a base catalyst can cause saponification, lowering the yield.

Past questions: 2016M-Q5, 2016S-Q28(b), 2019M-Q11, 2019M-Q13, 2020E-Q40(a), 2021E-Q39(a), 2024E-Q37(a)(ii)